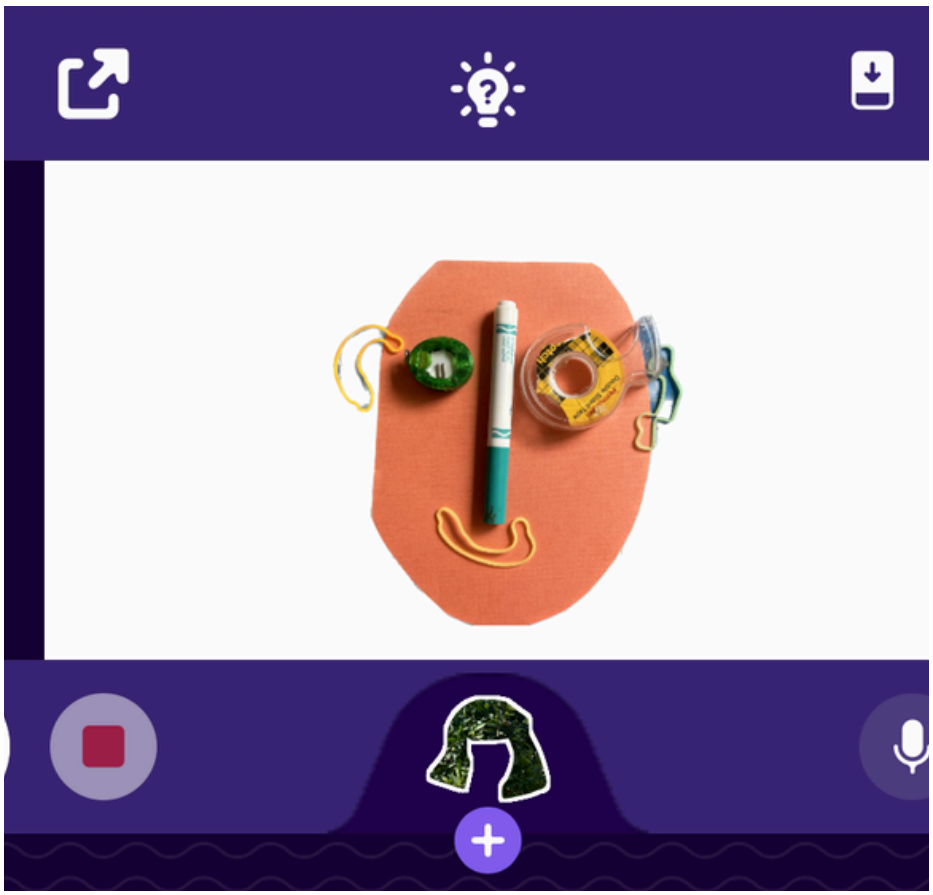
robots meet ARTS



Cofinançat per la Unió Europea

Número de projecte: 2023-1-FR01-KA220-SCH-000151881

BALLA JUGANT A FER CARES AMB OCTO STUDIO

L.A.B UCLL twa STIMMULI for social change UNIVERSITAT ROVIRA i VIRGILI de ark basisschool CC BY SA

Subjet:
Coding

Content:
How to create an animation with Octo Studio. Important message: This lesson should be taught after completing the first Octo Studio lesson.

Computational practices:

- Decomposing
- Pattern recognition
- Abstraction
- Algorithms
- Problem solving

Unplugged/plugged:
Plugged

Preparation time:
2 hours

Age:
8-12 years

Duration:
2 hours

Location:
Classroom

Group distribution:
Whole-class teaching and pairteaching

Difficulty 2/3

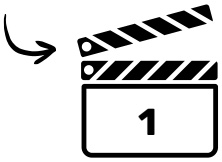
After this lesson, the children will be able to...

- Create a background using Octo Studio.
- Design a simple interactive program in Octo Studio combining movement, sound, text, and sensor input.
- Understand the meaning of the word 'pseudocode'.
- Work effectively in pairs by listening to each other's ideas and dividing tasks fairly.
- Communicate clearly with a partner to plan, build, and test a program together.
- Practice problem-solving by debugging and improving their program.
- Show creativity in designing characters (sprites), backgrounds, and interactions.

Materials needed:

For the teacher

- 1 x projector and cable to project from a tablet
- 1 x attachment with screenvideo's of the app. These symbols in this manual refer to a different video in the attachment.



For every pair

- 1 x scenario
- 1 x charged tablet with Octo Studio app (free)
- 1 x white A3-paper
- 1 x white A4-paper
- 1 x role cards (laminated)
- 1 x challenge cards
- 1 x whiteboardmarker
- 4 x post-it
- 1 x pencil

Accessible for all pairs

- a materials library with smaller materials such as: buttons, coins, clothespins, stretchers, caps in different sizes and colours.
- a materials library with larger materials such as: roll of sticky tape, fork, toys and other playfull parts that can ad personality and spark metaphorical expression.

Note: All materials are ready to print and can be found at the end of this manual.

Helpful links:

Octo Studio reference guide

- https://octostudio.org/resources/Reference%20Guide.pdf

Inspiration source for this lesson

- https://docs.google.com/presentation/d/e/2PACX-1vRuuU6uQNuqVtodR9bVa0IfI1Xm0Ec4FSo04BPEisfwGi551g9bSaKTfp1zgalKkVdbVITxNtYB3hLJ/pub?start=true&loop=true&delayms=60000&slide=id.p

Preparation and teacher notes

Download the software on the tablets and choose the correct language for your classroom.

Prepare the materials as listed above.

Optional: Divide the group already into pairs.

Links:

- To Hanock Piven's website: https://www.pivenworld.com/
- To inspiration video of making faces: https://vimeo.com/508166355/87a701bc4c

If there aren't enough tablets for each pair, consider using a microteaching approach: work with a small group while the remaining students engage with other classroom activities independently.

Meeting Hanoch Piven (5' whole class teaching)

Show Hanoch Piven's website and explore his works. Discuss with the children how he constructs his works: he makes faces out of everyday objects. Tell students that this this lesson, they will create and animate their own Hannoch Piven artwork using Octo Studio.

Observe theatrical expressions (10' in pairs)

Tell students that today, they will work on the four emotions happy, angry, scared and sad. Taking turns, they are going to observe each other with an emotion. Then discuss some tips for forming this emotion. The typical characteristics per emotion can be found below:

😊 Happy: Key facial features

- **Smile**
- **Bright, open eyes**
- **Raised eyebrows**
- **Rosy cheeks**

😢 Sad: Key facial features

- **Frown downward curved mouth**
- **Tears**
- **Lowered or slanted eyebrows**

😠 Angry: Key facial features

- **Furrowed eyebrows**
- **Narrowed or squinted eyes**
- **Tight, straight mouth**

😨 Scared: Key facial features:

- **Wide open eyes**
- **Raised eyebrows**
- **Small or open mouth in "O" shape**
- **Asymmetrical face**

Answer generated with ChatGPT

Making face expressions with the materials (10' in pairs)

Show the materials library. Explain that in a moment, students will make their own faces with the materials from the library on the white paper background. Let the pupils name the elements they can vary with: eyes, nose, mouth, eyebrows...

They are going to try to make all the 4 face expressions with the same materials. In this video you could find more inspiration.

Make a specific emotion (15' in pairs)

Hand out a scenario to each duo. They have to guess which emotion is behind the scenario. You can find the answers below. Then they will try to create that emotion with new or the same chosen objects from the library.

You get a test back and see that you scored 10/10. *(Happy)*
Someone pushes you off the swing on purpose while you're playing. *(Angry)*
You have to go to the dentist for a cavity, and you don't know if it will hurt. *(Scared)*
It's your birthday and everyone sings to you in class. *(Happy)*
You told the truth, but the teacher doesn't believe you. *(Angry)*
Your favorite stuffed animal is lost at school. *(Sad)*
You lose your parent in a crowded shopping center. *(Scared)*
You win a prize in a drawing contest at school. *(Happy)*
Your brother or sister takes your toy without asking and breaks it. *(Angry)*
Your pet is sick and has to go to the vet. *(Sad)*
You hear strange noises at night while you're alone in bed. *(Scared)*
You get to have an unexpected sleepover at your best friend's house. *(Happy)*

Creating the photoshoot set-up (10' whole class teaching)

Each duo makes the following setup:

- 1 white A3 sheet on the ground
- 1 white A4 sheet on top of the A3, with 5 cm folded over along the long side (see photo). This A4-sheet is the size of the frame. All the faces should fit inside this paper.

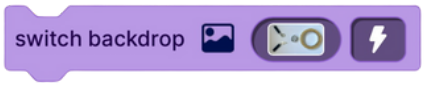
Divide the roles (5' in pairs)

Let the students think about the possible two roles, since there is only 1 tablet. Eg: one person operates the tablet and the onther gives instructions. Have them write both roles on the cards using a whiteboardmarker and let each student choose one role.

Creating a program for an animation (10' whole class teaching)

In the prior lesson the students learned how to program and work with sprites. In this lesson, they will learn how to program a background.

Open the app and start a new project, choosing a background as your starting point. For the background, take a picture that shows the first facial expression. As a sprite, add your name or signature to the project. Show the students how to add another background. Also show them this block:



Program the animation with the challenge card (20' in pairs)

Hand out the challenge cards with the pseudo-code. Let the kids find the program themselves. If they are ready, they can continue with challenge 2.

Challenge 1

1. I see my first background.
2. After one second I see my second background.
3. This program loops forever.

Optional: Creating scenario's (35' Whole class teaching)

Each pair writes their own scenario on a post-it note. Then, they create additional scenarios for the other emotions as well, each one on a separate post-it. Next, the children visit each other's work. They first read the different scenarios, and then watch the animation. Finally, they guess which scenario matches the animation

Clean up (10')

List of annexes

AT_1: Role cards

AT_2: Scenario’s

AT_3: Video 1

AT_4: Video 2

AT_5: Video 3

AT_6: Challenge cards

AT_7: Challenge cards solutions

AT_1: Role cards

Role 1:

Role 2:

AT_2: Scenarios

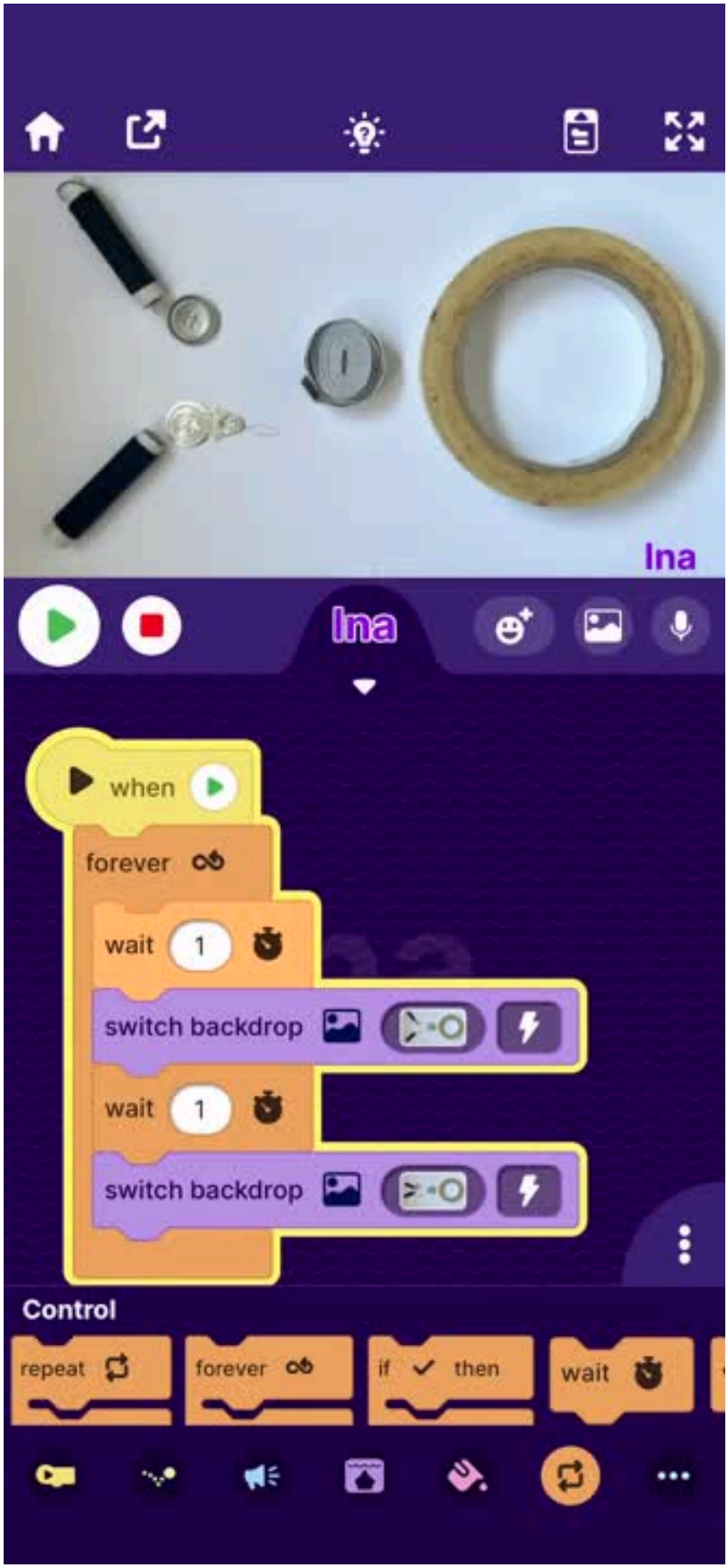
Scenario
You get a test back and see that you scored 10/10.
Someone pushes you off the swing on purpose while you're playing.
You have to go to the dentist for a cavity, and you don’t know if it will hurt.
It’s your birthday and everyone sings to you in class.
You told the truth, but the teacher doesn’t believe you.
Your favorite stuffed animal is lost at school.
You lose your parent in a crowded shopping center.
You win a prize in a drawing contest at school.
Your brother or sister takes your toy without asking and breaks it.
Your pet is sick and has to go to the vet.
You hear strange noises at night while you’re alone in bed.
You get to have an unexpected sleepover at your best friend’s house.

1

Creating a programme

Adding a background

Example programma



AT_6: challenge cards

Challenge cards

Octo Studio

Challenge 1

2. After one second I see my second background.

3. This program loops forever.

Challenge 2

Experiment:

Level 1: Add a soundeffect with every backgroundchange

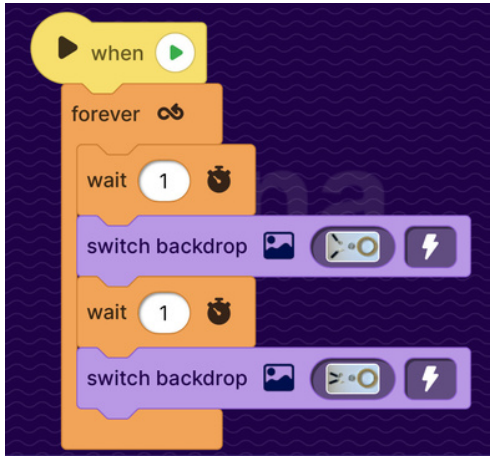
Level 2: Use a different start button (Eg: Every time I shake my device my background changes)

AT_7: challenge cards solutions

Challenge cards

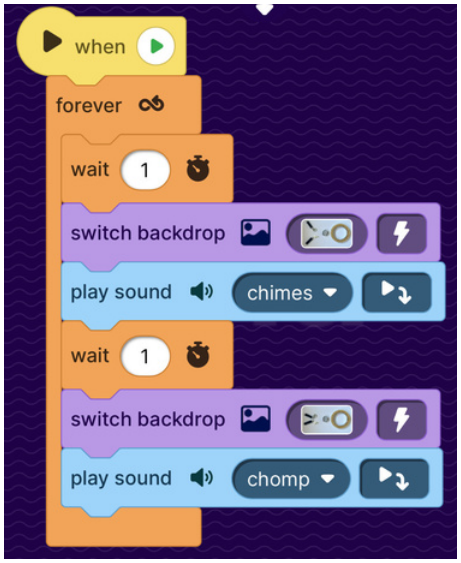
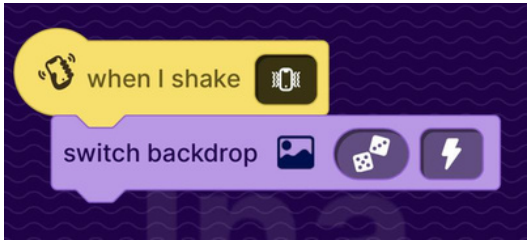
Octo Studio

Challenge 1



Challenge 2

Level 1:



Level 2: